

The Silent Tax: Typo-Robustness Analysis (GSM8K_FIX-SPELLCHECK_20000) DeepSeek-R1-Distill-Qwen-7B

All tables are *answered-only*: traces that hit the token cap before writing a final answer are excluded from the metrics. Configs are named `typo{rate}_real{ratio}`: *rate* = % of words corrupted, *real* = % of typos that form real words. Baseline is the clean input.

Accuracy and Flips

Table 1: Accuracy per config. *n* = questions; *n_answered* = questions that produced a final answer (not truncated at the token cap). *acc_answered* = correct / *n_answered*; *acc_strict* = correct / *n* (truncated counted as wrong); *ci_lo*/*ci_hi* = 95% bootstrap CI on *acc_strict*.

config	n	n_answered	acc_answered	acc_strict	ci_lo	ci_hi
clean	500	500	0.94	0.94	0.918	0.96
typo25_real10	500	491	0.8697	0.854	0.822	0.884
typo25_real40	500	489	0.8609	0.842	0.81	0.874
typo25_real70	500	489	0.8528	0.834	0.8	0.866
typo50_real10	500	490	0.7898	0.774	0.736	0.814
typo50_real40	500	481	0.7505	0.722	0.68	0.762
typo50_real70	500	492	0.7317	0.72	0.678	0.76
typo75_real10	500	481	0.7526	0.724	0.686	0.762
typo75_real40	500	474	0.6814	0.646	0.606	0.688
typo75_real70	500	466	0.6223	0.58	0.538	0.626

Table 2: Answered-only flips vs clean, paired by question over *n_both* (answered in both conditions). *r2w* / *w2r* = questions correct on clean but wrong on this config / wrong on clean but correct on this config; *r2w_ratio* = *r2w* / *n_both*, *w2r_ratio* = *w2r* / *n_both*; *net_flip_ratio* = *r2w_ratio* - *w2r_ratio* = (*r2w* - *w2r*) / *n_both*, the net rate at which this config worsens answers relative to clean.

config	n_both	r2w	w2r	r2w_ratio	w2r_ratio	net_flip_ratio	mcnemar_b	mcnemar_c	mcnemar_stat	mcnemar_p
typo25_real10	491	39	4	0.0794	0.0081	0.0713	39	4	26.884	2.2e-07
typo25_real40	489	45	6	0.092	0.0123	0.0798	45	6	28.314	1e-07
typo25_real70	489	49	6	0.1002	0.0123	0.0879	49	6	32.073	1e-08
typo50_real10	490	78	4	0.1592	0.0082	0.151	78	4	64.988	0.0
typo50_real40	481	98	8	0.2037	0.0166	0.1871	98	8	74.726	0.0
typo50_real70	492	110	8	0.2236	0.0163	0.2073	110	8	86.449	0.0
typo75_real10	481	100	9	0.2079	0.0187	0.1892	100	9	74.312	0.0
typo75_real40	474	129	6	0.2722	0.0127	0.2595	129	6	110.252	0.0
typo75_real70	466	153	5	0.3283	0.0107	0.3176	153	5	136.766	0.0

McNemar test (used in the flips table above).

- **What it is:** a paired significance test for before/after binary outcomes. It looks only at the questions that *changed* (the discordant pairs: $b = \text{right} \rightarrow \text{wrong}$, $c = \text{wrong} \rightarrow \text{right}$) and ignores those that stayed the same.
- **Null hypothesis:** typos break correct answers as often as they fix wrong ones ($b = c$, i.e. no net effect).
- **Statistic:** $(|b - c| - 1)^2 / (b + c)$ (column `mcnemar_stat`); `mcnemar_p` is its chi-square p -value with 1 degree of freedom.
- **When we reject it:** when `mcnemar_p` < 0.05 — the $\text{right} \rightarrow \text{wrong}$ vs $\text{wrong} \rightarrow \text{right}$ imbalance is too large to be chance.

Reasoning Length

Table 3: Absolute generated tokens per config (answered-only).

config	n_answered	mean_tok	median_tok	p90_tok	max_tok
clean	500	1523.6	1148.0	2750	15738
typo25_real10	491	1873.6	1310	3748	15664
typo25_real40	489	1856.0	1293	3593	11817
typo25_real70	489	1853.3	1291	3805	16241
typo50_real10	490	1975.3	1345.0	4071	12630
typo50_real40	481	2030.1	1313	4371	12962
typo50_real70	492	2204.4	1324.0	5072	14117
typo75_real10	481	2136.3	1409	4791	12836
typo75_real40	474	2396.9	1513.5	5548	17532
typo75_real70	466	2394.2	1572.0	5403	15496

Table 4: Mean generated tokens for correct vs wrong answers. `wrong_over_correct` = `tok_wrong` / `tok_correct` (mean tokens of wrong answers divided by mean tokens of correct answers).

config	tok_correct	tok_wrong	wrong_over_correct
clean	1361.4	4065.4	2.9862
typo25_real10	1592.2	3750.9	2.3558
typo25_real40	1543.3	3791.5	2.4567
typo25_real70	1544.5	3641.7	2.3579
typo50_real10	1601.4	3380.2	2.1107
typo50_real40	1592.6	3346.3	2.1012
typo50_real70	1659.5	3690.6	2.2239
typo75_real10	1680.0	3524.2	2.0977
typo75_real40	1739.0	3804.4	2.1877
typo75_real70	1787.6	3393.7	1.8985

Self-doubt

The self-doubt markers are curated words grouped into two *banks*, each matched as a case-insensitive whole-word pattern in the reasoning text. Density is matches per 1000 reasoning words.

- **second_guess** — the proposal’s canonical self-correction words plus direct reversals: *wait*, *hmm*, *actually*, *reconsider*, *but wait*, *hold on*, *on second thought*, *let me recheck*, *double-check*.

- **uncertainty** — hedging / “not certain” words: *maybe, perhaps, possibly, might be, could be, not sure, I guess, presumably, or maybe / alternatively, unclear / confused.*

Example reasoning: “*Wait, maybe the total is 12... hmm, actually let me recheck*” contributes *wait, hmm, actually* (second_guess) and *maybe* (uncertainty). A separate *typo-noticing* bank (*typo, misspelled, doesn’t make sense, ...*) is treated as repair behaviour, not self-doubt.

Table 5: Doubt-marker density per config: second_guess and uncertainty, per 1k words and per trace.

config	n.answered	second_guess_per_1k	second_guess_per_trace	uncertainty_per_1k	uncertainty_per_trace	total_per_1k	total_per_trace
clean	500	13.371	11.144	7.065	5.888	20.435	17.032
typo25_real10	491	14.123	15.224	13.186	14.214	27.308	29.438
typo25_real40	489	14.676	15.505	13.938	14.726	28.614	30.231
typo25_real70	489	14.148	14.914	13.427	14.153	27.575	29.067
typo50_real10	490	14.682	16.804	16.888	19.329	31.57	36.133
typo50_real40	481	14.545	16.911	16.125	18.748	30.67	35.659
typo50_real70	492	14.321	18.628	19.685	25.606	34.005	44.234
typo75_real10	481	14.616	18.208	19.758	24.613	34.375	42.821
typo75_real40	474	14.51	20.291	21.375	29.892	35.885	50.184
typo75_real70	466	13.865	19.56	22.05	31.107	35.916	50.667

Table 6: Per-marker density (markers per 1000 reasoning words): clean vs pooled-typo. discrimination = typo minus clean (larger = more typo-responsive). All bank markers, ranked by discrimination.

marker	category	clean_per_1k	typo_per_1k	discrimination
maybe	uncertainty	3.213	9.469	6.256
wait	second_guess	6.157	8.363	2.206
or maybe/alt	uncertainty	1.934	4.108	2.174
perhaps	uncertainty	0.828	1.989	1.161
unclear/confused	uncertainty	0.545	1.189	0.644
not sure	uncertainty	0.06	0.245	0.185
might be	uncertainty	0.115	0.282	0.167
could be	uncertainty	0.086	0.175	0.088
not entirely	uncertainty	0.017	0.055	0.038
I assume	uncertainty	0.019	0.049	0.03
presumably	uncertainty	0.0	0.011	0.011
actually	second_guess	0.319	0.33	0.011
possibly	uncertainty	0.005	0.007	0.002
reconsider	second_guess	0.0	0.0	0.0
on second thought	second_guess	0.0	0.0	0.0
but wait	second_guess	0.744	0.658	-0.086
hold on	second_guess	0.715	0.629	-0.087
wait/actually no	second_guess	0.48	0.391	-0.089
I guess	uncertainty	0.242	0.106	-0.136
hmm	second_guess	3.127	2.932	-0.195
let me recheck	second_guess	0.84	0.526	-0.314
double-check	second_guess	0.989	0.555	-0.434

Table 7: Doubt density (markers per 1000 reasoning words) split by final outcome, per category (2g = second.guess, un = uncertainty, tot = both). For a group, density = total markers in that group / total reasoning words in that group x 1000. *_correct = over correct-answer traces; *_wrong = over wrong-answer traces; *_wrong_over_correct = _wrong / _correct.

config	2g_correct	2g_wrong	2g_wrong_over_correct	un_correct	un_wrong	un_wrong_over_correct	tot_correct	tot_wrong	tot_wrong_over_correct
clean	12.927	15.532	1.201	6.085	11.839	1.946	19.012	27.371	1.44
typo25_real10	13.997	14.452	1.033	11.553	17.446	1.51	25.55	31.899	1.249
typo25_real40	13.78	16.765	1.217	11.573	19.455	1.681	25.353	36.22	1.429
typo25_real70	13.701	15.165	1.107	11.769	17.197	1.461	25.47	32.362	1.271
typo50_real10	13.586	16.473	1.212	13.437	22.525	1.676	27.023	38.998	1.443
typo50_real40	13.677	15.736	1.151	12.041	21.729	1.805	25.718	37.464	1.457
typo50_real70	13.595	15.141	1.114	14.793	25.215	1.705	28.388	40.357	1.422
typo75_real10	14.395	14.908	1.036	15.552	25.291	1.626	29.947	40.198	1.342
typo75_real40	14.112	14.877	1.054	15.768	26.55	1.684	29.881	41.427	1.386
typo75_real70	13.656	14.038	1.028	17.885	25.499	1.426	31.541	39.537	1.253

LLM judge. The proposal specifies a second, non-lexical measure for both dimensions: a separate model with a fixed, strict prompt. The judge is a different model from the one under test, called at temperature 0 on a seeded random subsample of answered traces per config. It is shown the corrupted-word list (*original* $\rightarrow$ *as shown*), the question, and the reasoning trace (head+tail window). It returns two scalar ratings and evidence: *self_doubt_score* from 0–10, where 0 means no second-guessing and 10 means pervasive uncertainty or looping; and *repair_understanding_score* from 0–5, where 0 means the question’s meaning is fully understood and 5 means the model is fundamentally lost because of the corruption. The judge also returns a one-line justification and verbatim evidence quotes; quotes are checked against the reasoning trace. The judge is a subsample measure, while the lexical tables remain the full-data result.

Repair Behaviour

Table 8: How each corrupted word was handled, per config (categories defined below the table).

config	n_corrupt_words	silent_fix_n	flagged_n	misread_n	not_used_n	silent_fix_frac	flagged_frac	misread_frac	not_used_frac
typo25_real10	3598	2977	159	34	428	0.8274	0.0442	0.0094	0.119
typo25_real40	3601	2760	315	93	433	0.7665	0.0875	0.0258	0.1202
typo25_real70	3587	2513	545	136	393	0.7006	0.1519	0.0379	0.1096
typo50_real10	7066	5794	358	79	835	0.82	0.0507	0.0112	0.1182
typo50_real40	6927	5071	777	192	887	0.7321	0.1122	0.0277	0.128
typo50_real70	7055	4585	1249	307	914	0.6499	0.177	0.0435	0.1296
typo75_real10	9720	7710	645	184	1181	0.7932	0.0664	0.0189	0.1215
typo75_real40	9508	6583	1332	369	1224	0.6924	0.1401	0.0388	0.1287
typo75_real70	9213	5354	2019	583	1257	0.5811	0.2191	0.0633	0.1364

A *corrupted word* is a word that received a typo — it differs between the clean and typo question (e.g. *sum* $\rightarrow$ *sun*). We recover these by diffing the two questions, then place each into one bucket by which form appears in the reasoning:

- **silent_fix** — only the original word (*sum*) appears: a quiet recovery.
- **flagged** — both forms appear (*sum* and *sun*): noticed and fixed.
- **misread** — only the corrupted form (*sun*) appears: read as the wrong word.
- **not_used** — neither form appears in the reasoning.

Columns *_n are counts; *_frac are the share of n_corrupt_words.

Table 9: Word handling pooled by real-word ratio: the non-word vs real-word contrast.

real_ratio	n_corrupt_words	silent_fix_frac	flagged_frac	misread_frac	not_used_frac
10	20384	0.8085	0.057	0.0146	0.1199
40	20036	0.7194	0.121	0.0326	0.127
70	19855	0.6271	0.192	0.0517	0.1291

Table 10: Misread rate split by final correctness. A typo'd word is 'misread' when the reasoning uses only its corrupted form (e.g. sun instead of sum). Within each outcome group, misread rate = (typo'd words that were misread) / (all typo'd words examined), pooled over the group. misread_correct = over problems answered correctly; misread_wrong = over problems answered wrong; wrong_over_correct = misread_wrong / misread_correct.

config	misread_correct	misread_wrong	wrong_over_correct
typo25_real10	0.0081	0.0171	2.103
typo25_real40	0.0232	0.0411	1.776
typo25_real70	0.033	0.0632	1.918
typo50_real10	0.01	0.0153	1.534
typo50_real40	0.0255	0.0337	1.32
typo50_real70	0.0414	0.0485	1.171
typo75_real10	0.0167	0.0256	1.535
typo75_real40	0.0333	0.0497	1.491
typo75_real70	0.0534	0.0778	1.457

Real-word Effect

This section isolates the effect of typo *kind* (real-word vs non-word), holding the question, corrupted positions, and rate fixed — something the vs-clean accuracy section cannot do. Three views:

- a **controlled paired** comparison of low-real vs high-real variants on the *same* questions, with a McNemar test (paired table);
- a **per-typo logistic regression** comparing the marginal harm of one real-word vs one non-word typo (logit table);
- whether wrong answers **flag the corruption** more or less as the real-word share rises (silent-failure table).

Read the signs, deltas and p -values in the tables below for this dataset's result.

Table 11: Controlled paired comparison of two TYPO variants against EACH OTHER (baseline is the low-real variant, NOT clean – so this is not the vs-clean accuracy table). Within a fixed rate the same questions are corrupted in the same positions, only the typo KIND differs. Restricted to questions answered in both variants. $\text{acc_low_real} / \text{acc_high_real}$ = accuracy on the lower / higher real-word variant; $\text{delta_acc} = \text{acc_high_real} - \text{acc_low_real}$; r2w = questions correct in the low-real variant but wrong in the high-real one; w2r = the reverse; mcnemar_p = McNemar test on that paired change. Negative delta with $\text{r2w} \neq \text{w2r}$ means real-word typos are more harmful than the non-word typos they replaced.

rate	compare	n_both	acc_low_real	acc_high_real	delta_acc	r2w	w2r	mcnemar_p
25	real10_to_real70	481	0.869	0.8565	-0.0125	36	30	0.53825266
25	real10_to_real40	481	0.8732	0.8628	-0.0104	37	32	0.63013033
25	real40_to_real70	481	0.8607	0.8565	-0.0042	27	25	0.88970694
50	real10_to_real70	483	0.795	0.7371	-0.058	69	41	0.01004289
50	real10_to_real40	472	0.7945	0.75	-0.0445	63	42	0.05096194
50	real40_to_real70	475	0.7516	0.7453	-0.0063	45	42	0.83021759
75	real10_to_real70	451	0.7517	0.6186	-0.133	103	43	1.05e-06
75	real10_to_real40	458	0.7664	0.6878	-0.0786	81	45	0.00182053
75	real40_to_real70	446	0.6883	0.6323	-0.0561	68	43	0.02272782

Table 12: Logistic regression $\text{correct} \sim \text{num_real} + \text{num_nonword}$ over all answered typo traces ($\text{num_real} / \text{num_nonword}$ = count of real-word / non-word typos in the problem). coef_^* = change in log-odds of a correct answer per one added typo of that kind (more negative = more harmful); $\text{odds_}^* = \exp(\text{coef})$; intercept is the model constant. n = number of traces fit.

n	coef_num_real	coef_num_nonword	odds_real	odds_nonword	intercept
4353	-0.06858	-0.02849	0.9337	0.9719	1.92604

Table 13: Among WRONG answers only, does the model flag the corruption, by real ratio? $\text{explicit_notice_frac}$ = fraction of wrong-answer traces whose reasoning uses strong typo-flagging language (typo / misspelled / doesn't make sense / ...); $\text{silent_frac} = 1 - \text{explicit_notice_frac}$. The proposal predicts real-word failures are quieter ($\text{explicit_notice_frac}$ should fall as real rises).

real_ratio	n_wrong	explicit_notice_frac	silent_frac
10	286	0.6958	0.3042
40	339	0.7139	0.2861
70	380	0.7289	0.2711

Lexical Grid (overview)

Table 14: All four marker families (second_guess, uncertainty, typo_noticing, repair_words) across configs.

config	second_guess	uncertainty	typo_noticing	repair_words
clean	13.371	7.065	1.281	0.969
typo25_real10	14.123	13.186	3.331	2.443
typo25_real40	14.676	13.938	3.383	2.859
typo25_real70	14.148	13.427	3.455	2.444
typo50_real10	14.682	16.888	3.803	2.935
typo50_real40	14.545	16.125	3.852	2.922
typo50_real70	14.321	19.685	4.063	3.37
typo75_real10	14.616	19.758	4.673	3.904
typo75_real40	14.51	21.375	4.34	3.459
typo75_real70	13.865	22.05	4.614	3.731

Spellcheck Recovery